Mohamed Helmy

Software Engineer | Flutter Developer

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PROFILE

Flutter Developer with strong expertise in building scalable cross-platform mobile applications using Flutter and Dart. Experienced in Clean Architecture, MVVM, and Bloc/Cubit state management. Skilled in integrating REST APIs, Firebase services, and AI-powered features into production-ready mobile applications. Passionate about developing high-performance mobile apps and solving real-world problems using modern software engineering practices.

EDUCATION

Bachelor of computer science 2020 – 2024

Benha University - Faculty of Computers and Artificial Intelligence Benha, Egypt

Computer Science major,

GPA: 3.6,

Graduation Project: Excellent

PROFESSIONAL EXPERIENCE

Software Engineer

- **Software Engineer (Military Service):** Dec 2024 - Dec 2025, Cairo, Egypt

  Developed a secure Flutter-based system for digitizing military educational logistics and lecture tracking.

  Key Contributions:
  - Built a Flutter application to automate attendance and reporting systems.
  - Developed an offline-first synchronization system to ensure reliability in restricted network environments.
  - Automated reporting workflows, improving operational efficiency and data accuracy.
  - Implemented secure data handling for sensitive internal systems.

- **Flutter Developer (Freelancer):** Dec 2024 - Present
  - Developed cross-platform mobile applications using Flutter and Dart.
  - Designed scalable app architecture using Clean Architecture and MVVM.
  - Integrated REST APIs and Firebase services.
  - Implemented Bloc/Cubit state management for efficient UI updates

SKILLS

Programming Languages
Dart, Java, Kotlin, Python, C++, C, SQL

Mobile Development
Material UI, Responsive UI, Flutter Widgets, Navigation, Async Programming

Architecture & Patterns
Clean Architecture, MVVM, Repository Pattern, SOLID Principles, OOP, Design Patterns

State Management
Bloc, Cubit, Provider

Backend & APIs
REST APIs, JSON Parsing, Firebase Authentication,
Firebase Firestore, Firebase Realtime Database

Database
Hive, MySQL, Oracle

Tools
Git, GitHub, Dio, HTTP, Postman, Android Studio, VS Code

Additional Skills
Machine Learning, CNN, Computer Vision, AI Models, Embedded Systems, IoT Development, Unit Testing, Widget Testing, Data Structure, Algorithms,

PROJECTS

ScanCare - AI Medical Diagnosis App Graduation Project Dec 2023 - June 2024

Technologies: Flutter*, Python, CNN, Firebase, Bloc, Google Maps API, Stripe*

- Developed an AI-powered medical application that analyzes medical images using Convolutional Neural Networks (CNN).
- Integrated Google Maps API to locate nearby clinics and hospital.
- Implemented secure payment processing using Stripe.
- Built scalable architecture using MVVM and Bloc.
- Integrated 5 AI models, automating 70% of tasks.

E-Commerce Mobile Application Dec 2023 - Jan 2024

Technologies*: Flutter, Firebase, MVVM, REST APIs*

- Developed a Flutter-based e-commerce mobile application with features including:
  - Product browsing
  - Cart management
  - Favorites system
  - Order history
- Implemented advanced state management using Bloc architecture.

Bookly - Book Discovery App Dec 2023 - Feb 2024

Technologies: Flutter, Bloc, REST APIs

- Developed a mobile application to search and browse books using public APIs.
- Implemented advanced state management using Bloc architecture.
- Built using the MVVM (Model-View-View Model) architectural pattern

Mobile Store Management Application | MVVM Sept 2023 - Oct 2023

Technologies*: Dart, MVVM, Bloc State Management, Firebase*

- *Developed a comprehensive mobile store management application providing distinct feature sets for administrators and customers.*
- *For Administrators: Implemented full CRUD (Create, Read, Update, Delete) functionality for managing product inventory, viewing detailed user records, and managing invoice processing.*
- *For Customers: Enabled users to browse the latest product offerings, manage purchases, and utilize a dedicated maintenance request section.*
- *Structured the application using the MVVM (Model-View-ViewModel) architectural pattern and Bloc State Management to ensure scalability and predictable state control.*
- *Utilized Firebase for robust authentication, real-time data synchronization, and reliable backend service delivery*

IoT Smart Home System May 2024 - June 202

Technologies*:* Arduino*,* Python*,* Firebase*,* Sensors

- Built an IoT dashboard to remotely control home appliances.
- Integrated sensors for:
  - temperature
  - humidity
  - fire detection
  - gas leakage
- Implemented voice control for smart home automation.

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INVOLVEMENT & INTERNSHIPS

Android developer July 2022 - Oct 2022
ITCRC Benha, Egypt

Mobile development using (Flutter) July 2023 - Oct 2023
ITCRC Benha, Egypt

Flutter developer Jan/2023 - Apr/2023
Information Technology Institute– ITI Menoufia, Egypt

Community & Content

@eng_mo723

Technical Content Creator – YouTube

Created 100+ educational videos about:

- Software engineering
- Embedded systems
- Assembly programming
- Google Cloud

SOFT SKILLS

Communication | Teamwork | Adaptability | Problem Solving | Time Management | Leadership